Hobit – Project Proposal

Habit Tracking and Reminder Mobile Application

1. About the Project

Hobit is a mobile application designed to help users build and maintain healthy daily routines. The application allows users to create personal habits, schedule reminders, record daily completions, and monitor their progress. The primary target audience is university students who want to organize daily activities and maintain consistent routines through a simple and user-friendly mobile application.

2. Project Objectives

- Make habit creation and management simple and streamlined.
- Help users maintain consistent daily routines over time.
- Provide customizable reminders for planned habits.
- Record daily habit completion status accurately.
- Visualize progress through intuitive calendar and statistics features.
- Support offline usage with reliable cloud synchronization.

3. Planned Features

- **User Authentication:** Email/password login, Google sign-in integration, and password recovery.
- **Habit Management:** Create, edit, set starting dates, configure frequencies (daily, weekly, one-time), and archive habits.
- **Daily Tracking:** View daily lists, mark completions, and store dated records.
- **Reminders:** Custom reminder times, local notifications, and scheduling controls.
- **Calendar & Stats:** Interactive completion calendar, streak tracking, and weekly analytics.
- **Synchronization:** Local storage for offline access with Firebase Cloud Firestore sync.

4. Application Screens & Workflow

The planned application includes the Login Screen, Home Screen, Add/Edit Habit Screen, Habit Details Screen, Calendar Screen, Statistics Screen, and Settings Screen. The habit creation workflow is structured into two intuitive steps: entering the habit name and starting date first, followed by configuring repetition frequency and reminder times.

5. Technology Stack

Technology / Tool	Purpose
Flutter	Cross-platform mobile application development framework.
Dart	Application programming language.
Provider & ChangeNotifier	State management and UI state change notifications.
Inter (Typeface)	Clean and readable primary typography family supporting visual hierarchy.
Firebase Authentication	User authentication (Email/password and Google Sign-In).
Cloud Firestore	Cloud data storage and synchronization for online states.
Local Database & Notifications	Offline data access, local record storage, and scheduled reminders.

6. Software Architecture

Hobit follows the Model-View-ViewModel (MVVM) architectural pattern to ensure clean separation of concerns:

- **View:** Flutter screens, form components, and card layouts designed with a soft green-based color palette (#2C6658 primary, #E9F1E9 secondary, #EDF4EE background, #FCFCFA form cards, #F3F7F1 input fields) for a calm user experience.
- **ViewModel:** Manages screen state, input validation, loading states, and the two-step creation workflow.
- **Model & Repository:** Represents application entities (User, Habit, Completion, SyncMetadata) and coordinates data operations between local storage and Firestore.

7. Project Team & Responsibilities

Team Member	Main Responsibilities
Rana Girgin	Flutter interface, navigation, forms, calendar, and statistics.
Leyla Gülnur Gür	Data models, local database, Firebase authentication, Firestore sync, and iOS integration.
Duygu Eroğlu	Habit completion logic, notification scheduling, and Android integration.

8. Development Status & Future Improvements

Status: Currently in project planning and interface design stage, defining application features, technical architecture, and UI wireframes.
Future Improvements: Detailed progress insights, enhanced habit suggestions, customized themes, and optimized synchronization.

Note: Developed for academic purposes as part of the MTM4681 Mobile Programming course.